Why Observability is Critical for Kubernetes

The importance of observability in today's AI-driven world cannot be underestimated, especially when it comes to addressing the challenges that Kubernetes can face.

The introduction of cloud computing around 2006 transformed infrastructure management. It enabled organisations to provision and manage resources on demand without the need to build their own physical infrastructure. Docker continued this evolution of infrastructure by introducing containers to enable application portability. Applications rely on specific libraries, files, and configurations that do not always work consistently across different operating systems. Containers solved this by packaging applications with their dependencies, allowing them to run reliably across different environments.

Unlike virtual machines, which require a complete operating system, containers run as lightweight processes, consuming fewer resources. This enables hundreds of containers to run on a single machine and allows applications to start within seconds instead of the longer boot times required by virtual machines. However, as container adoption grew, the next challenge was managing containers at scale.

This is where Kubernetes plays a role. It manages containers across different applications using deployments, StatefulSets, and other management tools. This makes it possible to efficiently deploy, coordinate, and manage large-scale containerized environments.

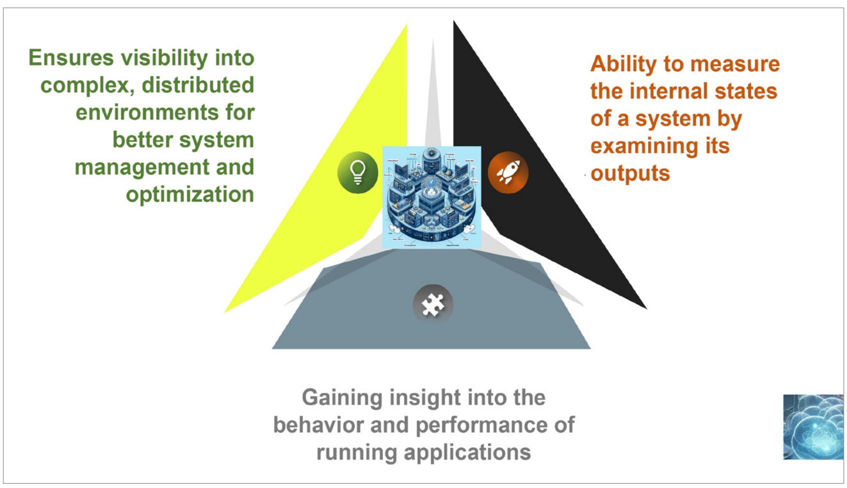
Saurabh Mishra, DevOps lead, Global Payments

Why observability is needed in modern applications

As applications become more complex, challenges arise during production. When systems exhibit errors or unexpected failures, operations teams often don't have a deep understanding of the application codebase. This makes troubleshooting slower and affects service reliability and customer experience. To close this gap between development and operations, organisations adopt DevOps, which combines development skills with operational knowledge.

With the rise of AI technologies such as ChatGPT and Gemini after 2021, modern systems have become even more distributed and complex, increasing the need for deeper visibility into application behaviour.

Observability addresses this



Observability and its fundamentals